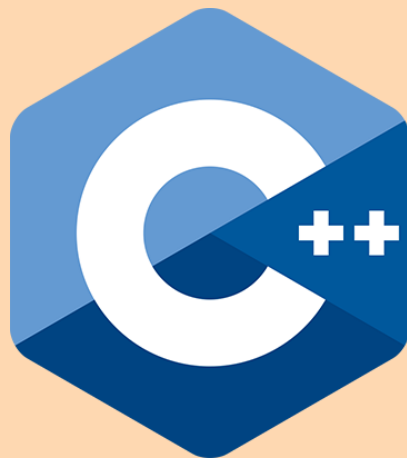


3. WHILE . . .

[Hengfeng Wei \(魏恒峰\).](#)
hfwei@hnu.edu.cn



Review

If Statement

Switch Statement

Logical Expressions

Overview

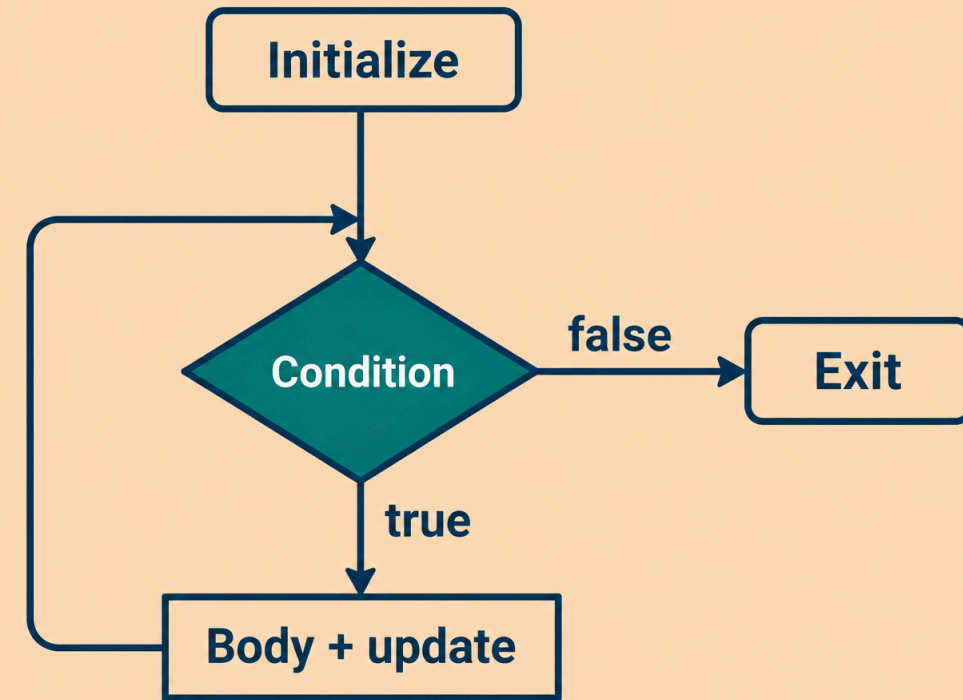
While (Do-While) Statement

break Statement

while Statement

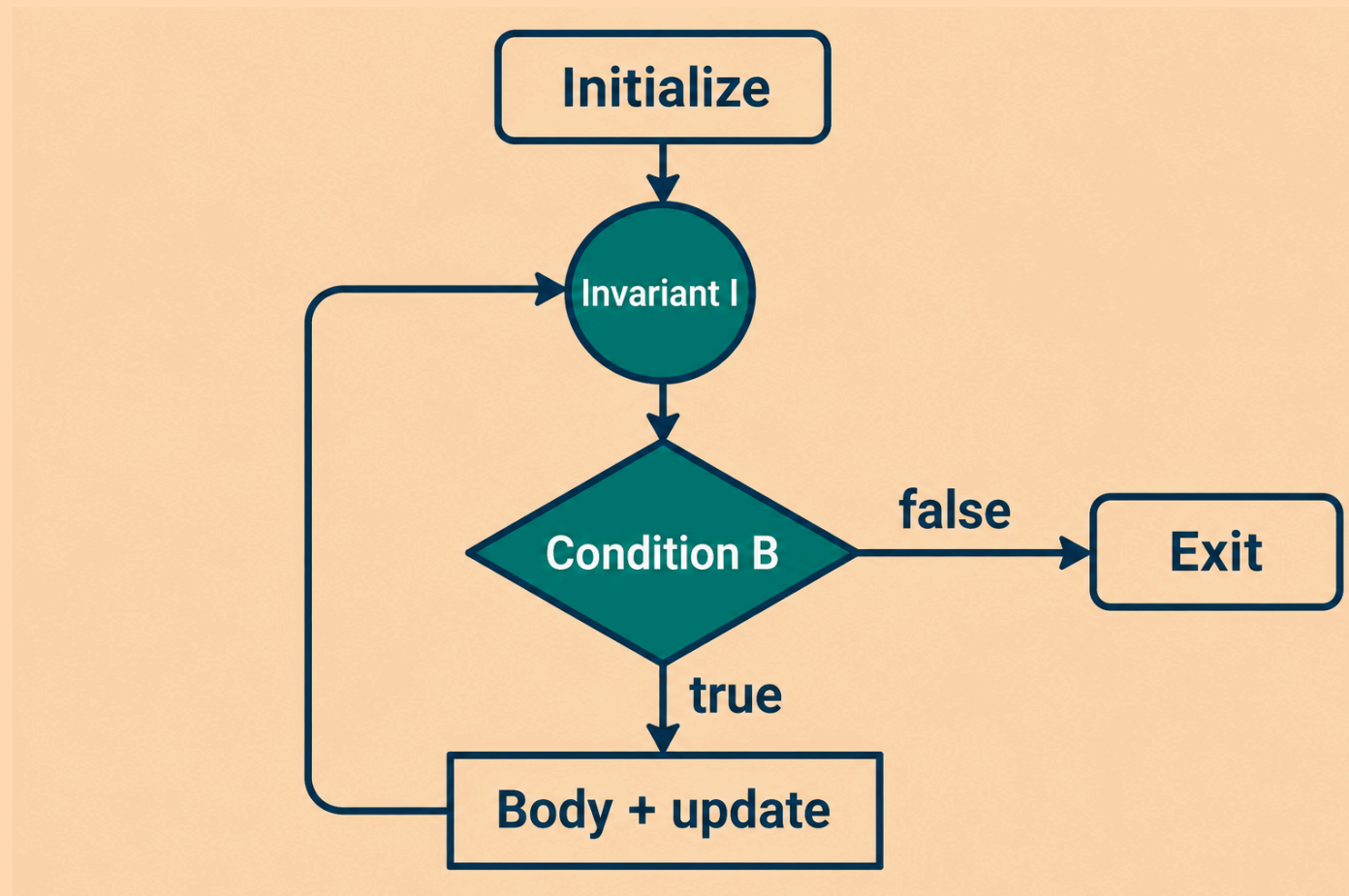
```
// initialize  
while (condition) {  
    // body + update  
}
```

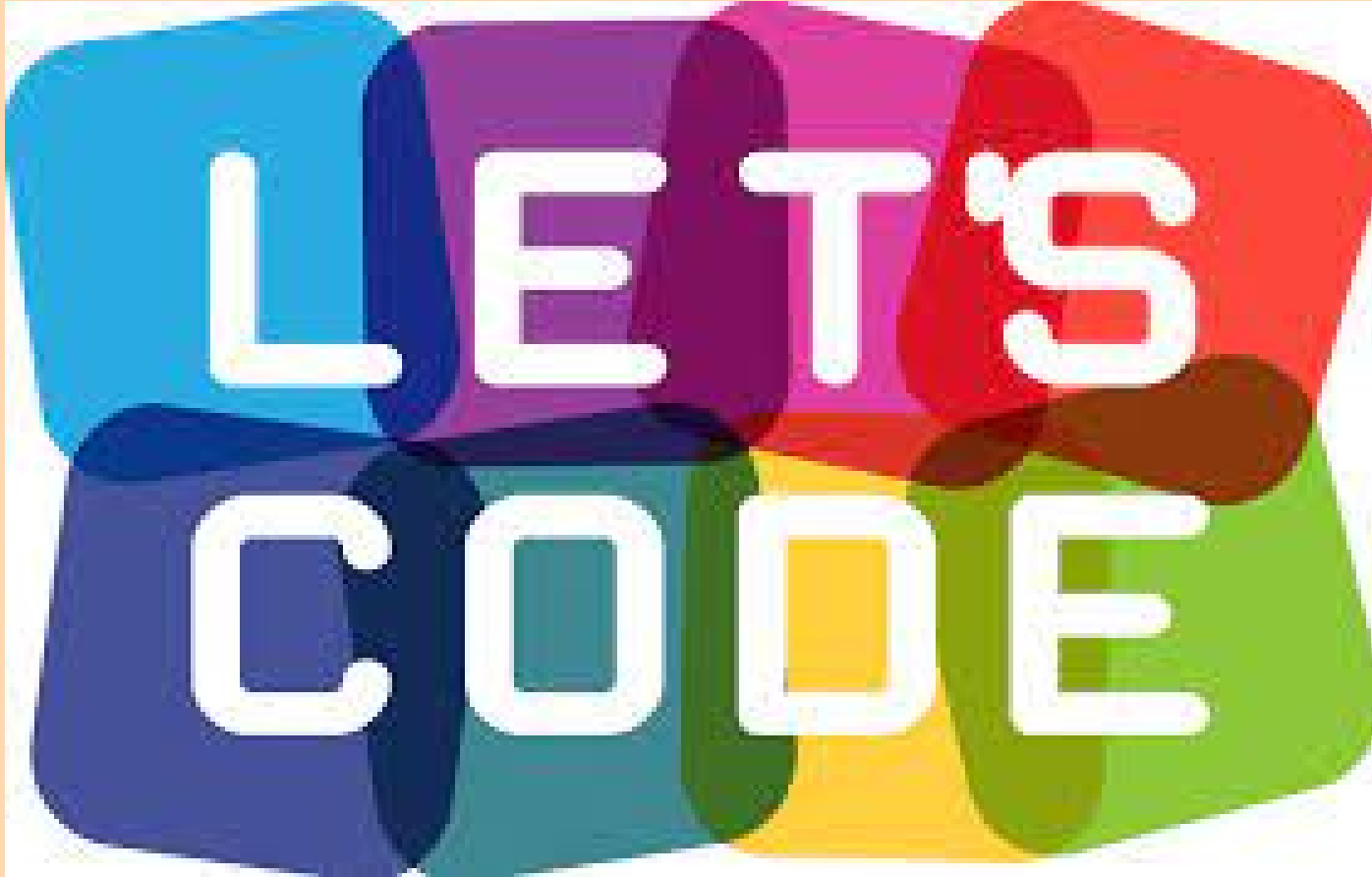
May execute **zero** times.



Loop Invariant as Abstraction

What remains true before every iteration?





euclid.cpp fib.cpp digits.cpp
palindrome.cpp binary-search.cpp

Greatest Common Divisor

$$\gcd(a, b) = \gcd(b, a \% b)$$



$$(48, 18) \longrightarrow (18, 12) \longrightarrow (12, 6) \longrightarrow (6, 0)$$

Loop Invariant

$a_0, b_0 > 0$: **original inputs**

$$\boxed{\gcd(a, b) = \gcd(a_0, b_0)}$$

$$(a, b) \xrightarrow{b \neq 0} (b, a \% b)$$

Fibonacci Sequence

$$F_0 = 0$$

$$F_1 = 1$$

$$F_n = F_{n-1} + F_{n-2} \quad (n > 1)$$

Loop Invariant

$i : \text{index}, \quad 0 \leq i \leq n$

$\text{previous} = F_i$	$\text{current} = F_{i+1}$
-------------------------	----------------------------

$\underbrace{(F_i, F_{i+1})}_{\text{before}} \longrightarrow \underbrace{(F_{i+1}, F_i + F_{i+1})}_{\text{after: } (F_{i+1}, F_{i+2})}$

Number of Digits



Identify the Place & Value
of Digits in a
Three-Digit Number



1 Hundred

127

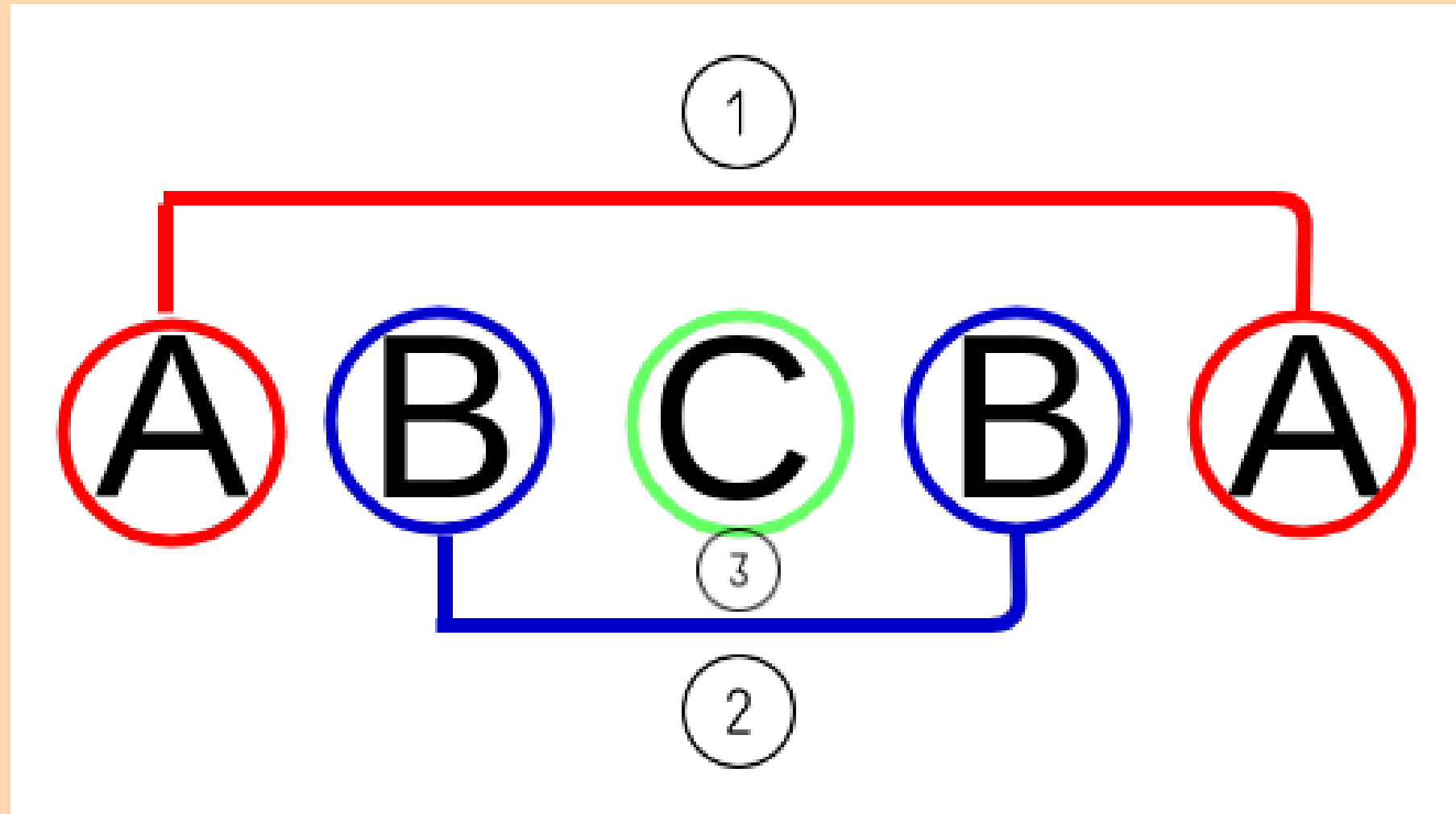


2 Tens

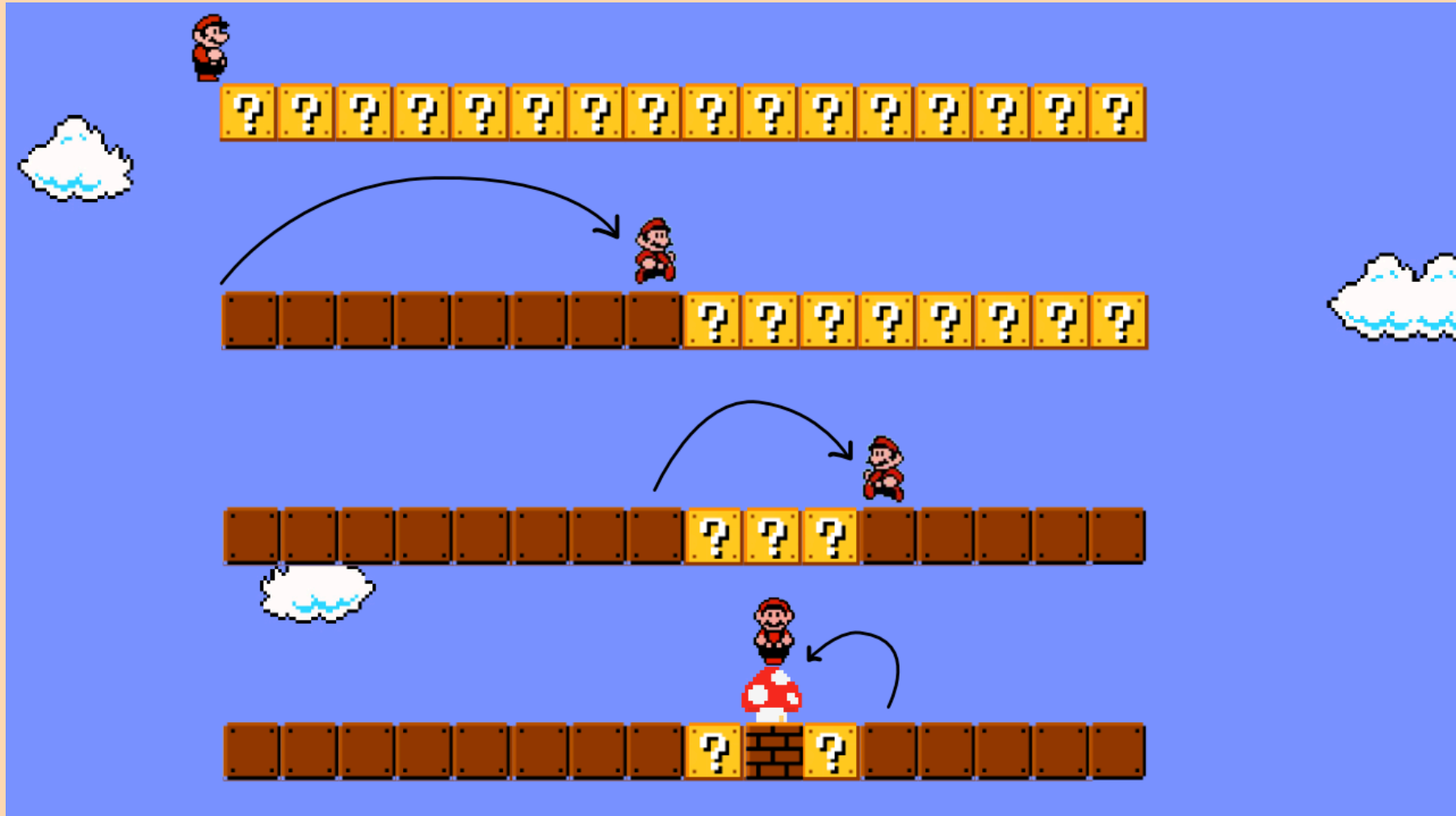


7 Ones

Palindrome



Binary Search



Array_INITIALIZER (DO)

- `int numbers[4] = {1};`

First element is `1`; the rest are `0`.

- `int numbers[] = {0, 1, 2};`

Size is deduced: `3`.

- `int numbers[4] = {};`

All elements are initialized to `0`.

Array Initializer (DON'T)

```
int numbers[4];
```

Initialize before reading.

Array Initializer (DON'T)

```
int numbers[];
```

Specify the size, or provide initial values.

```
int numbers[] = {0, 1, 2};
```

